MASTER IN ASTROPHYSICS AND SPACE SCIENCE

TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Optimisation & Monte Carlo Sampling

Introduction to Numerical Methods for Astrophysics

Fritz Ali Agildere

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Overview

Overview

Frequentist View

- *With Prior Knowledge:* out of all n different but equally likely possible outcomes of an event, there are k that have a property A which is therefore measured with probability

$$P(A) = \frac{k}{n}$$

- *Without Prior Knowledge:* out of n independent measurements the outcome A occurs k times, defining

$$P(A) = \lim_{n \to \infty} \frac{k}{n}$$

Bayesian View

- $P(A|B)$ quantifies the plausibility of an assumption A under known information given by B
- *Kolmogorov Axioms* underpin the rules for calculating with probabilities
- $P(A, B) = P(A)\,P(B|A)$ quantifies the probability of measuring A and B simultaneously, and from assuming $P(A, B) = P(B, A)$ we find

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Continuous Distributions

- *Cumulative Distribution Function:* $F(x) = P(X \leq x)$
- *Probability Density Function:* $f(x) = \frac{d}{dx}F(x)$

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx = F(b) - F(a)$$

- *Normalization:* $1 = \int_{-\infty}^{+\infty} f(x)\,dx \qquad \lim_{x\to-\infty} F(x) = 0 \qquad \lim_{x\to+\infty} F(x) = 1$

Expectation Values

$$E[h(x)] = \int_{-\infty}^{+\infty} h(x)\, f(x)\, dx$$

- *Mean* or first raw moment: $\mu = E[X] = \int_{-\infty}^{+\infty} x f(x)\, dx$
- *Variance* or second raw moment: $\sigma^2 = E\left[(X - E[X])^2\right] = \int_{-\infty}^{+\infty} \left(x - E[x]\right)^2 f(x)\, dx$

Overview

Drawing Numbers

$$x \sim U(a, b) \qquad\qquad y \sim N(\mu, \sigma)$$

- *Uniform Distribution:* $U(a, b) = f(x|a, b) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$
- *Gaussian Distribution:* $N(\mu, \sigma) = f(y|\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2\right)$

Number Generation

- *Pseudo Random Numbers:* $x_{j+1} = g(x_1, \dots, x_j)$
  - deterministic & reproducible
  - efficient & transparent
  - almost indistinguishable with long periods of recurrence
- *Equal Distribution Generators:*
  - *Linear Congruential*
  - *Linear Feedback Shift*
  - *Permutated Congruential*
- *Arbitrary Distributions:* obtained from uniformly distributed numbers by various methods

Overview

Motivation

- *Model Optimisation:* finding the best possible model parameters
  - maximising objective function *(Fitting)*
  - minimising loss function *(Machine Learning)*
- *Example:* modified blackbody spectrum for galactic dust emission (M. Juvela, *A&A* 2023)

$$I_\nu(T,\beta) \equiv \frac{\nu^{3+\beta}}{\exp\left(\frac{h\nu}{kT}\right) - 1}$$

Example

$$I_\nu(T,\beta) = \frac{\nu^{3+\beta}}{\exp\left(\frac{h\nu}{kT}\right) - 1}$$

- *Objective:* optimise $\theta \equiv (T, \beta)$ such that the weighted squared error is minimal
- *Assumptions:*
  - independent measurements generated by $I_\nu(\theta)$ with normally distributed uncertainty
  - infinitesimally small error in ν so that we only need σ_ν in I_ν direction
  - suppose we have no prior information on the parameter distribution

Example

- *Likelihood:*

$$-\log L(I_\nu|\nu,\sigma_\nu,\theta) \propto \chi^2(\theta) = \sum_{i=1}^{N} \frac{\left(I_{\nu,i} - I_\nu(\theta)\right)^2}{\sigma_{\nu,i}^2}$$

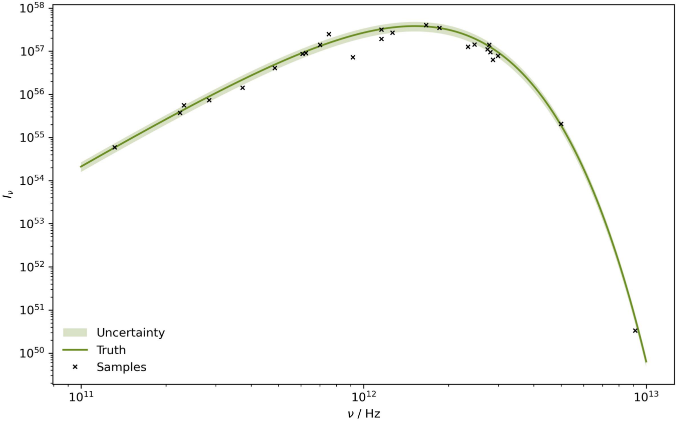
- *Posterior:*

$$\log P(\theta|I_\nu,\nu,\sigma_\nu) \propto -\chi^2(\theta)$$

- *Objective:* minimise the $\chi^2(\theta)$ measure to maximise the posterior probability

Measurements

How to we proceed from this data to maximise the likelihood for our nonlinear model?



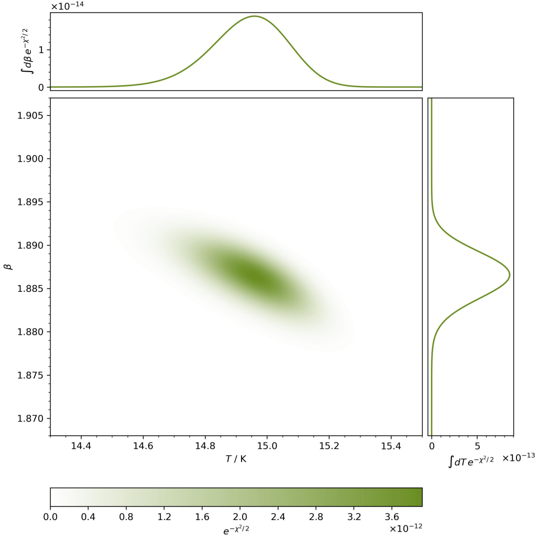
Overview

Parameter Grid

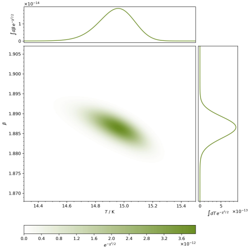
- *Idea:* in our $D = 2$ dimensional parameter space, we can create an $N \times N$ grid of (T, β) points and evaluate the likelihood at each one
- *Preparation:* for this as well as some of the following methods, it is a good idea to normalize the variation of each individual parameter
  - from the literature we expect $(T, \beta) \in [10, 25] \times [1.5, 2.0]$ as constraints
  - resize these intervals by defining $T \mapsto \tilde{T} = T / 30$ with $I_\nu(T, \beta) \to I_\nu(\tilde{T}, \beta)$
- *Problem:* as D increases we have N^D gridpoints, and with more complex parameter distributions, this procedure quickly becomes intractable due to computational limits

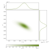
Parameter Grid

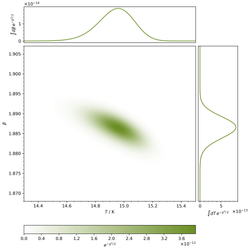
Can we think of more elegant ways to move even through high dimensional parameter spaces?

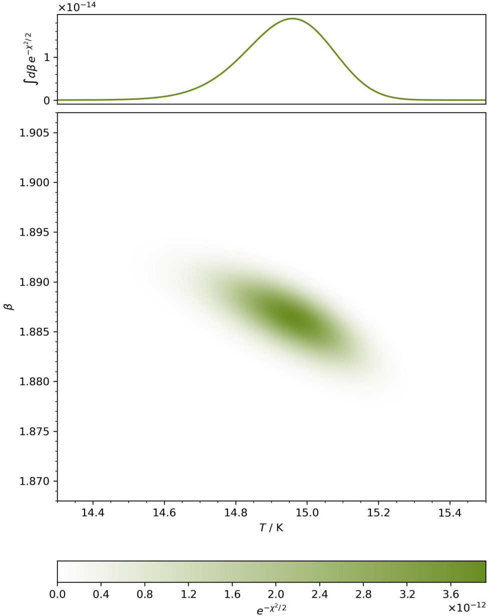


Where I Studied

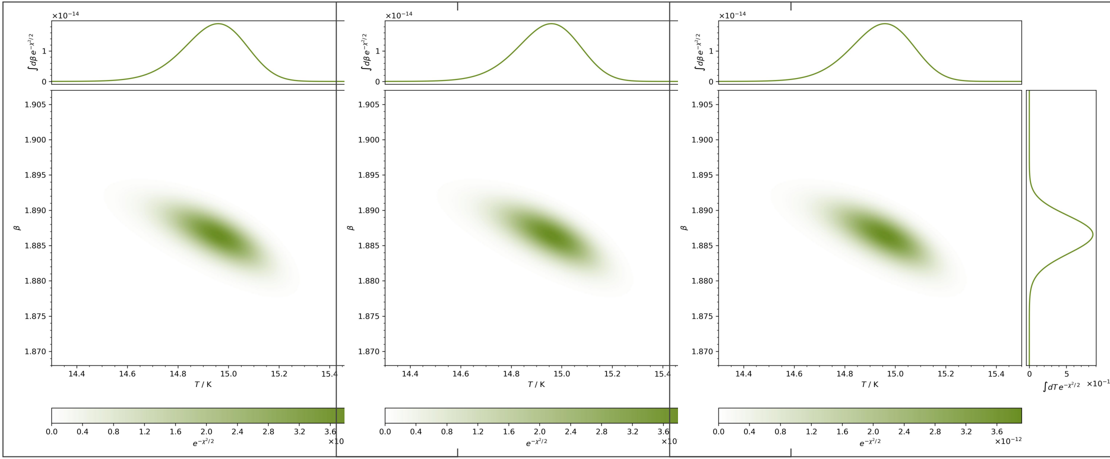






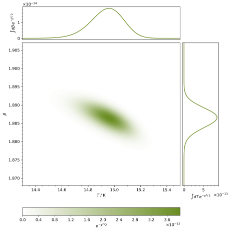


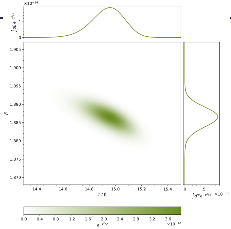
What I Studied

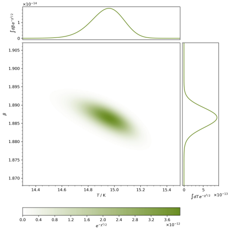


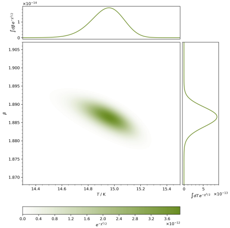
Why Astrophysics

- Interesting
  - Awesome Scales
  - Readily Observable
- Interdisciplinary
- International

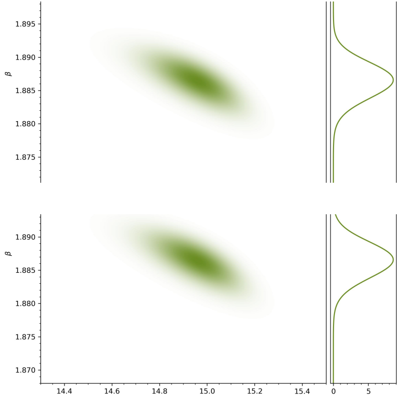


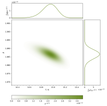


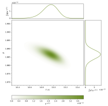


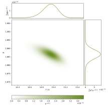


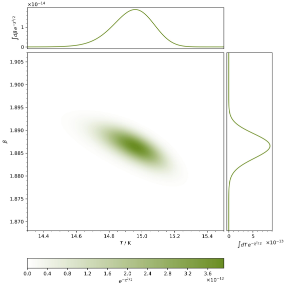
Where I Have Been



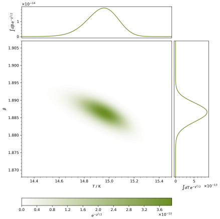


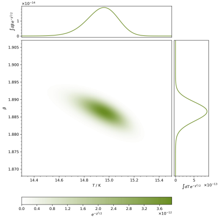


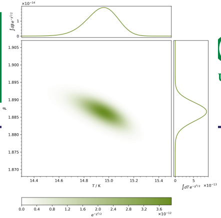


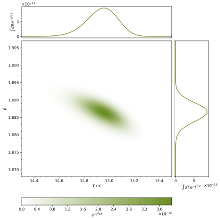


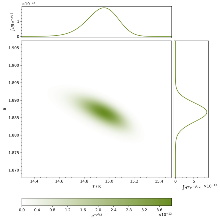
Other Interests











- Nature Photography
- Hiking & Cycling
- Programming
- Politics
- Meeting New People

Thank You!